

FENT2321

Wind Energy and Design of a Wind Turbine

Assignment 7: Generator Design to match the blades - System design

Markus Arntzen

Contents

1	Problem 1: Operating point variables	3
1.1	Generator geometry	3
1.2	Generator performance	3
1.2.1	Induced voltage at no load	3
1.2.2	Power curves	3
1.2.3	Operating condition parameters	4
1.2.4	Discussion	4

1 Problem 1: Operating point variables

1.1 Generator geometry

Poles	2
Phases	1
Coils	4
Length per coil	26.7m
Copper wire diameter	1.18mm
Number of turns per coil	60

1.2 Generator performance

1.2.1 Induced voltage at no load

RPM	E_A [V]
RPM _{design} - 300	51.6
RPM _{design}	63.6
RPM _{design} + 300	75.7

1.2.2 Power curves

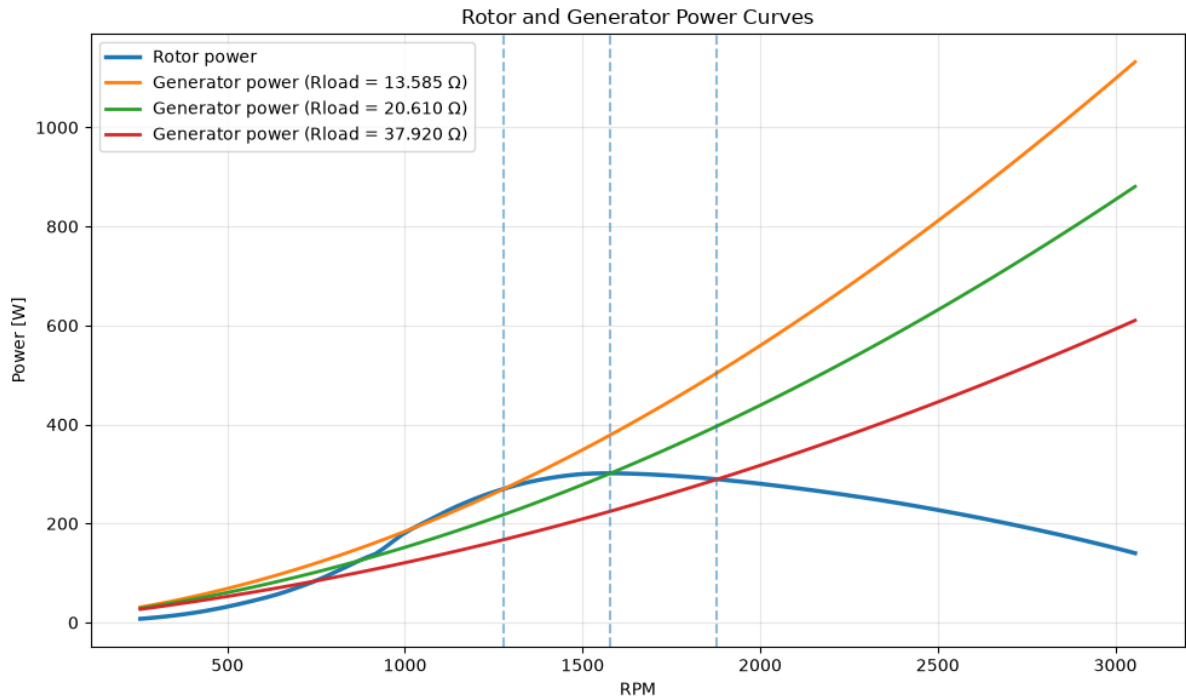


Figure 1: Power curves for different values of R_{load}

1.2.3 Operating condition parameters

RPM	R_L [Ω]	P_{el} [W]	T_{gen} [Nm]	R_A [Ω]	X_S [Ω]	L_S [H]	η_{gen}	η_{tot}
$RPM_{design} - 300$	16.465	128.58	1.823	1.877	1.968	0.014694	52.66%	52.66%
RPM_{design}	20.61	163.18	1.823	1.877	2.430	0.014694	54.13%	54.13%
$RPM_{design} + 300$	24.765	197.79	1.823	1.877	2.891	0.014694	55.14%	55.14%

1.2.4 Discussion

- When R_L is increased, the rotor of the turbine need to output more power. Since the torque the wind exerts on the turbine blades remain the same, the rotational speed need to increase to increase the power to the generator, thus increasing RPM.
- If the generator is unable to extract the maximum mechanical power produced by the rotor blades, the electrical power output will be limited by the generator's capacity. If the rotor produces more mechanical power than the generator can absorb, the rotor will accelerate due to the resulting positive torque imbalance. This acceleration may shift the tip speed ratio away from its optimal value, reducing the aerodynamic efficiency of the rotor. The turbine will therefore be unable to convert all the available wind energy into useful electrical power. The overall efficiency may decrease due to both the generator limitation and the possible deviation from the optimal operating point.
- When the wind speed is reduced to half its design value, the wind turbine will not necessarily remain at its optimal operating point. For our system, the optimal rotor speed would decrease from 1579 rpm to approximately 789.5 rpm to maintain the same tip speed ratio. Assuming the rotor maintains its optimal power coefficient, the mechanical power available at the design point of 301.47 W would decrease to approximately 37.68 W. The corresponding optimal generator torque would decrease from 1.823 Nm to approximately 0.456 Nm. However, since the generator load resistance is kept constant at 20.61 Ω , the generator torque may not match the optimal rotor torque at the reduced wind speed. The actual operating point is determined by the intersection of the rotor and generator torque curves. Therefore, the turbine will only operate optimally if this intersection occurs close to the required rotor speed of 789.5 rpm. Otherwise, the tip speed ratio will deviate from its optimal value, reducing the aerodynamic efficiency.